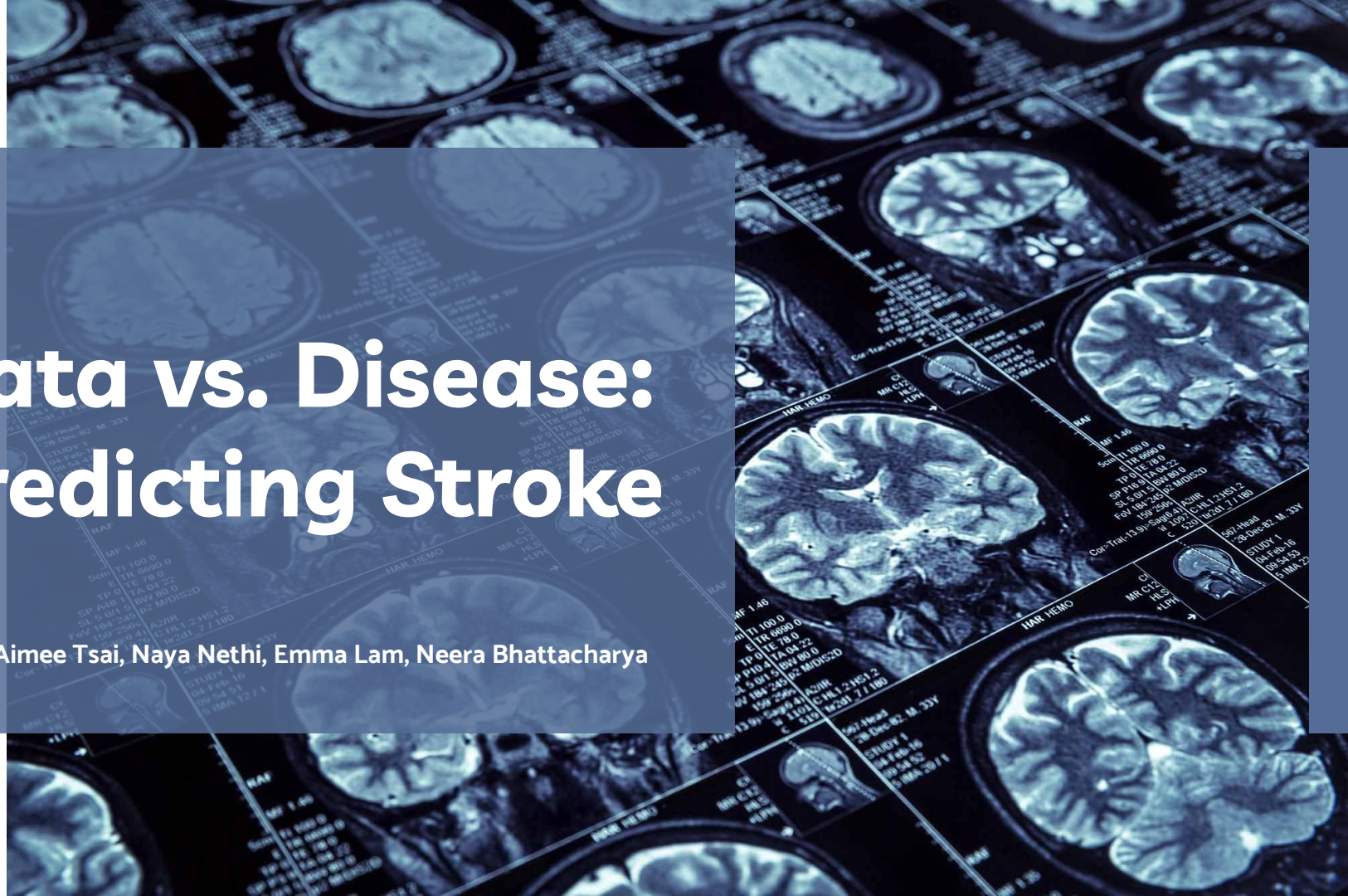
Data vs. Disease: Predicting Stroke

By: Aimee Tsai, Naya Nethi, Emma Lam, Neera Bhattacharya



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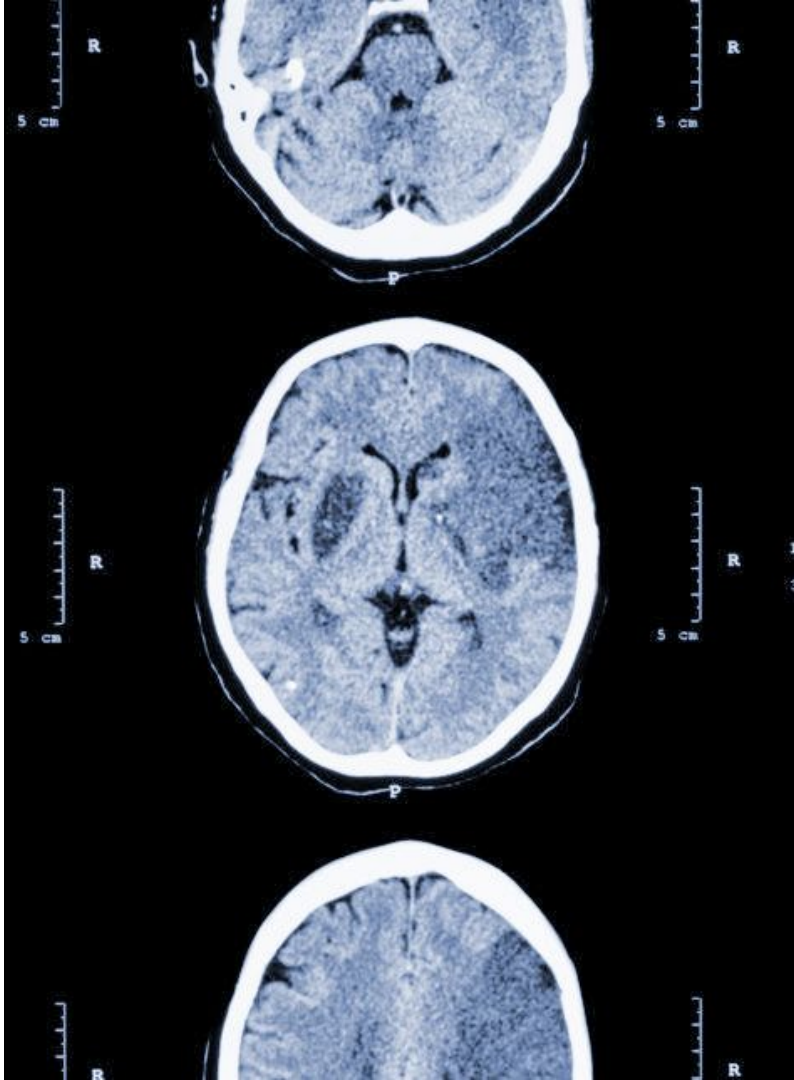
INTRODUCTION

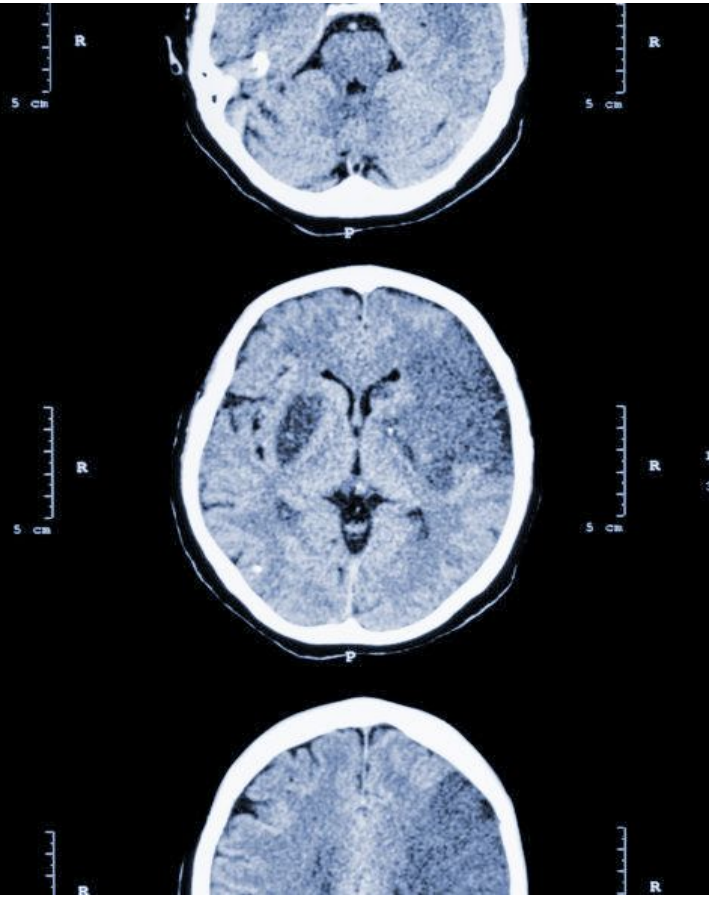
Strokes are one of the leading causes of deaths worldwide and the fifth leading cause of death in the U.S., so this project aims to build predictive models using **linear regression, model selection techniques**, and **LASSO regression** to determine the most important health variables associated with stroke risk in the United States. The predictor variables we are focusing on include: **obesity, smoking, bad physical health, bad mental health, asthma, diabetes, high blood pressure, sleep issues,** and **high cholesterol.**



WHY ARE WE RESEARCHING THIS?

Identifying the most significant risk factors in this study can enhance our understanding of stroke risk, leading to better public health strategies and improved patient outcomes.

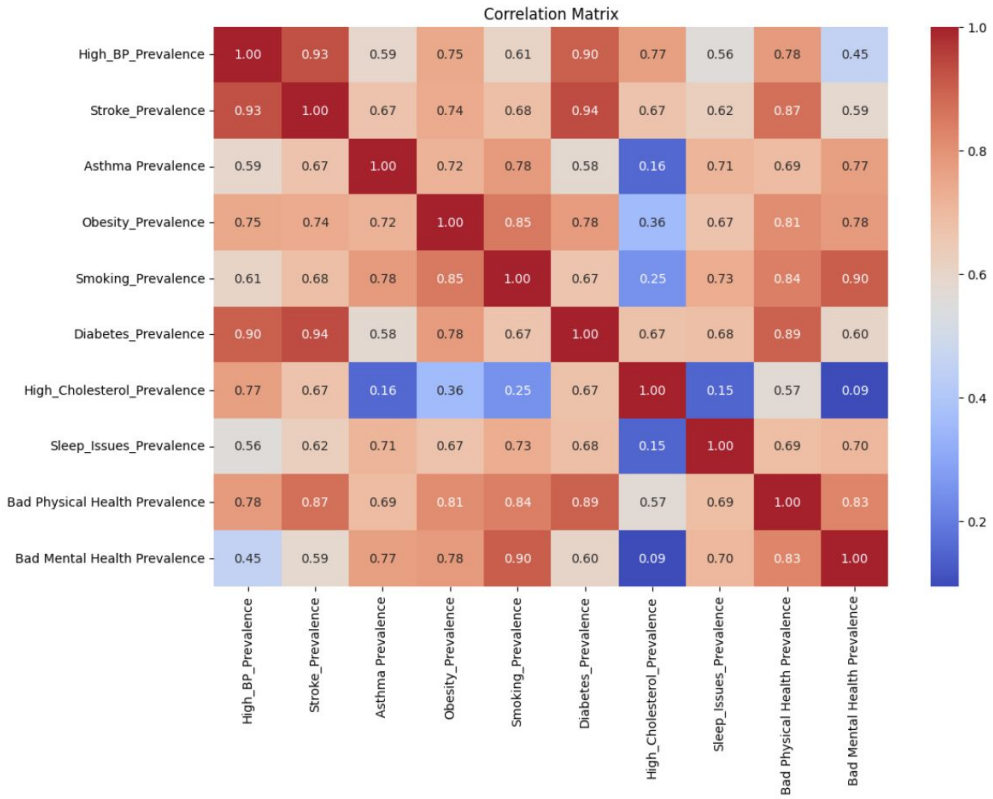
MOTIVATION



DATA

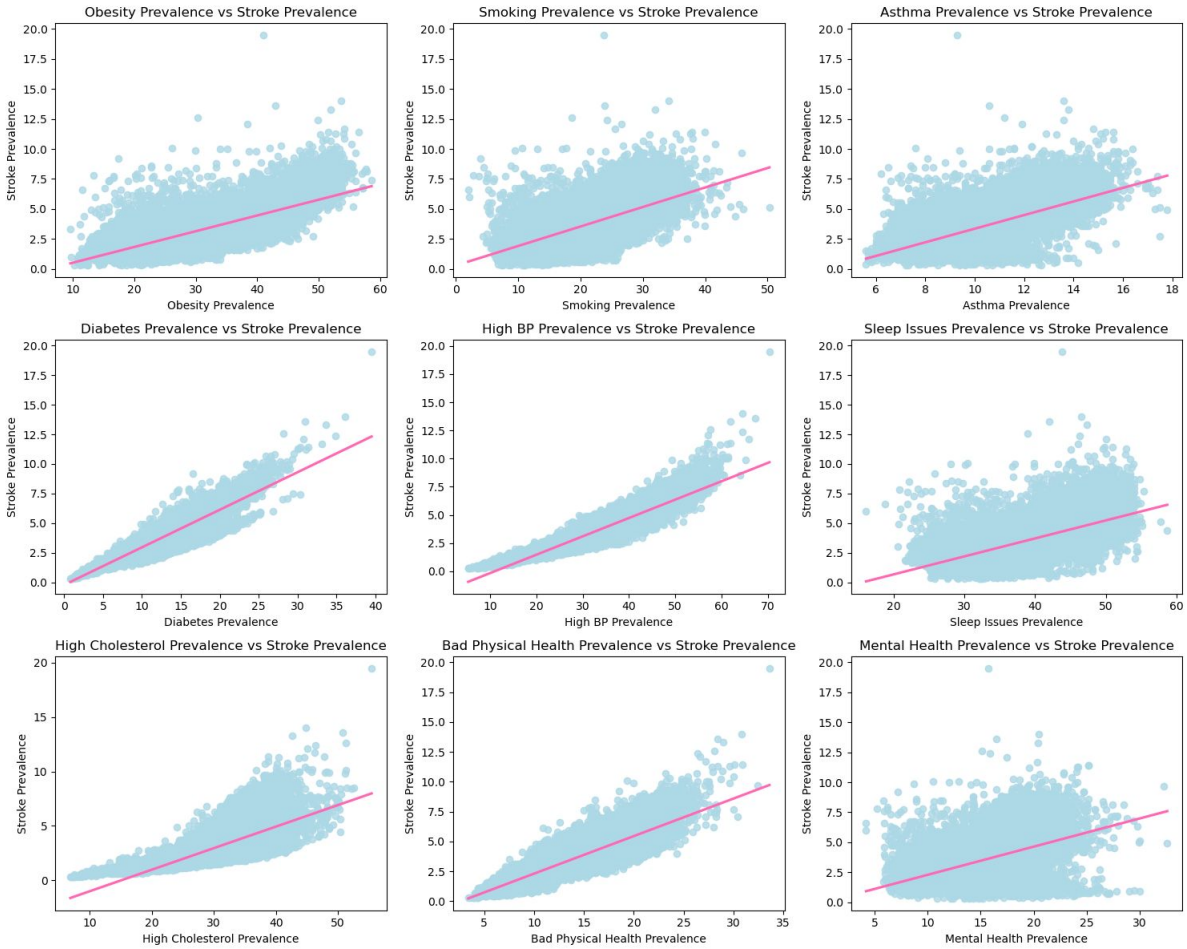
- CDC website U.S. Census, last updated on December 2023
- Predictor Variables: obesity, high blood pressure, diabetes, high cholesterol, asthma, sleep issues, bad physical health, bad mental health.
- To preprocess the data, we removed NA variables and filtered according to what variables we wanted to focus on. We also removed outliers.

Exploratory Data Analysis

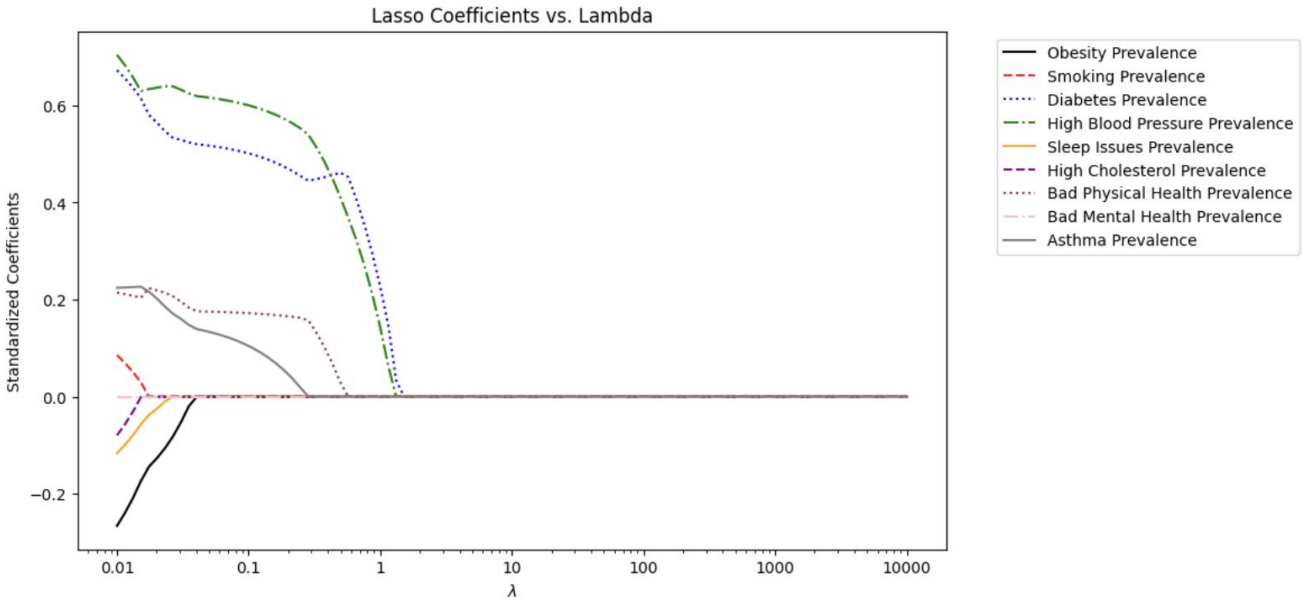


- From this correlation matrix, we can identify some of the **highly correlated variables:**
  - **Stroke** and **high blood pressure** (0.93)
  - **Stroke** and **diabetes** (0.94)
  - **Stroke** and **bad physical health** (0.87)
  - **Diabetes** and **bad physical health** (0.89)
  - **Smoking** and **bad mental health** (0.90)

EDA: Simple Linear Regression



Lasso Regression



The Lasso coefficients graph shows that **diabetes and high blood pressure** are **last few nonzero coefficients** at higher values of λ, representing the **strongest predictors** for stroke risk.

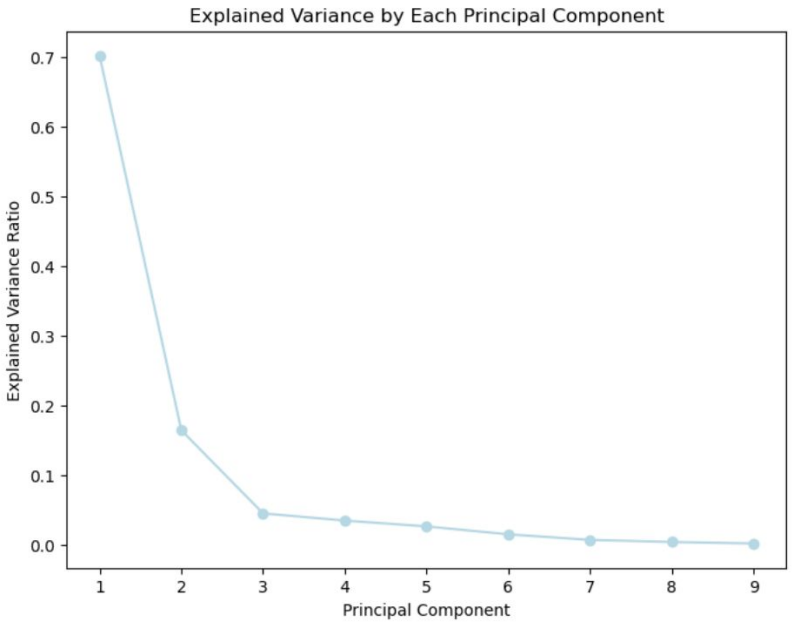
Multiple Linear Regression

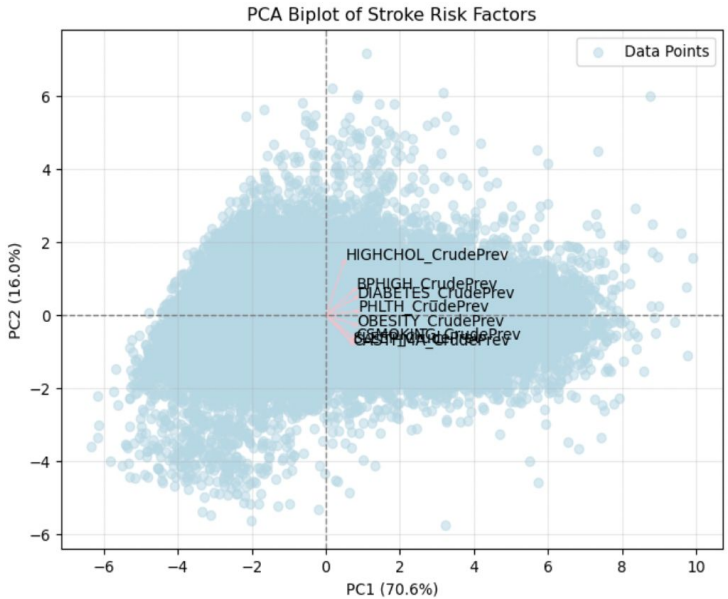
```
                            OLS Regression Results
==============================================================================
Dep. Variable:       STROKE_CrudePrev   R-squared:                       0.953
Model:                            OLS   Adj. R-squared:                  0.953
Method:                 Least Squares   F-statistic:                 6.101e+04
Date:                Mon, 10 Mar 2025   Prob (F-statistic):               0.00
Time:                        15:00:21   Log-Likelihood:                -7169.4
No. Observations:               27210   AIC:                         1.436e+04
Df Residuals:                   27200   BIC:                         1.444e+04
Df Model:                           9
Covariance Type:            nonrobust
==========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                    0.6833      0.036     19.179      0.000       0.613       0.753
BPHIGH_CrudePrev         0.0955      0.001    107.808      0.000       0.094       0.097
CASTHMA_CrudePrev        0.1510      0.002     65.163      0.000       0.146       0.156
OBESITY_CrudePrev       -0.0489      0.001    -81.102      0.000      -0.050      -0.048
CSMOKING_CrudePrev       0.0430      0.001     42.948      0.000       0.041       0.045
DIABETES_CrudePrev       0.1691      0.002     84.578      0.000       0.165       0.173
HIGHCHOL_CrudePrev      -0.0653      0.001    -61.628      0.000      -0.067      -0.063
SLEEP_CrudePrev         -0.0386      0.001    -60.972      0.000      -0.040      -0.037
PHLTH_CrudePrev          0.1306      0.003     48.379      0.000       0.125       0.136
MHLTH_CrudePrev         -0.0863      0.002    -38.480      0.000      -0.091      -0.082
==============================================================================
Omnibus:                     9189.988   Durbin-Watson:                   2.003
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           223886.120
Skew:                           1.072   Prob(JB):                         0.00
Kurtosis:                      16.888   Cond. No.                     1.36e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.36e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
Mean Squared Error: 0.10875126988348488
R-squared: 0.9499223928245386
```

Most variables behave as expected (e.g., **high blood pressure, diabetes, smoking, and asthma increase stroke prevalence**). However, some unexpected negative relationships (**obesity, high cholesterol, mental health issues**) suggest **potential multicollinearity.**

PCA





Principal Component Analysis reduces the complexity of the data by condensing the predictor variables into fewer and new variables which are called Principal Components. The first graph shows us that only three Principal Components are needed to explain over 90% of the variance in the data.

Forward Stepwise Selection

```
Selected Features: ['DIABETES_CrudePrev', 'BPHIGH_CrudePrev', 'CASTHMA_CrudePrev', 'OBESITY_CrudePrev', 'PHLTH_CrudePrev', 'HIGHCHOL_CrudePre
v', 'SLEEP_CrudePrev', 'CSMOKING_CrudePrev', 'MHLTH_CrudePrev']
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All of the predictor variables were selected, which means that they all contributed individually to improving the model, and none could be removed using Forward Stepwise selection.
In regards to the coefficients– the predictors are highly correlated, which means that the directions can change. Even though we would think that the predictors all contribute to higher stroke risk (isolated), when they are considered together, due to multicollinearity, some coefficients could appear negative.

Backward Stepwise Selection

```
Selected Features: ['BPHIGH_CrudePrev', 'CASTHMA_CrudePrev', 'OBESITY_CrudePrev', 'CSMOKING_CrudePrev', 'DIABETES_CrudePrev', 'HIGHCHOL_Crude
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The backward selection process also identified which health predictors most strongly associated with stroke risk. The retained variables—such as diabetes, high blood pressure, asthma, smoking, and poor physical health—align with known risk factors for stroke, reinforcing their significance. However, obesity and high cholesterol showed negative associations, which indicates complex interactions or multicollinearity. The model accounts for 95.3% of the variation in stroke risk, indicating that it effectively captures the main health-related factors.

MAIN RESULTS

- Based on our model selection, we found that all of our predictor variables are significant with p-values of under 0.05. This means there is strong evidence that all these factors have a statistically significant relationship with stroke, and they contribute to predicting stroke risk.
- Based on our **EDA** and **Lasso regression**, we found the two strongest predictor variables of stroke risk to be **diabetes** and **high blood pressure**. These findings align with existing medical research, as both conditions are well-documented risk factors for stroke due to their impact on cardiovascular health. For instance, diabetes can lead to damaged blood vessels and increased plaque buildup, while high blood pressure can strain arterial walls, making them more prone to blockage or rupture.

Discussion and Outlook

Strengths:

- Utilizing linear regression, model selection techniques, and LASSO regression allows for robust comparisons and helps identify the most important predictors while avoiding overfitting.
- Using CDC and U.S. Census data ensures credibility, as these sources are well-established for epidemiological studies.

Limitations:

- Data source only covers 500 U.S. cities and was last updated in 2023.
- Several other factors not included in this study can significantly impact stroke risk:
  - Age, sex, family medical history, diet, drug use, socioeconomic and environmental factors, biological factors

Potential Areas for Improvement:

- Incorporating Additional Variables
  - Merging external datasets together to include other variables that could improve model's explanatory power
- Creating interaction terms (e.g., obesity x diabetes) may help identify combined effects of risk factors.

Conclusion

After many regression models and selection techniques, we identified **diabetes and high blood pressure as the most influential predictors of stroke risk**, which correlates with existing medical research. While all of our chosen health variables significantly contributed to stroke prediction, multicollinearity introduced complexities in interpreting the coefficients. Our model accounts for 95.3% of the variance in stroke prevalence, demonstrating how effective it is. Future research could improve predictive accuracy by integrating additional factors like age, socioeconomic status, and lifestyle behaviors.